

Intro to Lasers

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Objectives

- Learn basic Inkscape design
- Importing a design for cutting
- Learn to operate under supervision by an authorized person
- Safe Operation



Requirements

- Must be supervised by authorized user
- Must use only during office hours or open-house.
- 18+ or supervised by parent/guardian
- \$20 Donation to Spark
 - Please help us, it takes time and effort to keep these machines in good condition.
 - We also have to pay for facilities, electricity, etc.
 - You will not find a better deal anywhere!



Next Steps

- If you become a member, there is additional training to:
 - Operate unsupervised
 - Maintain and align machine
- If you enjoy this, we encourage you to become a member and get actively involved in the community.



Machine Basics

- 150 Watt CO2 Laser (10.6um wavelength)
- Water cooled
- Adjustable bed height
- Usable Work Area: 550mm x 850mm (21.5in by 33.4in)
- Air extraction
- Air assist to improve cut quality/move debris away from nozzle

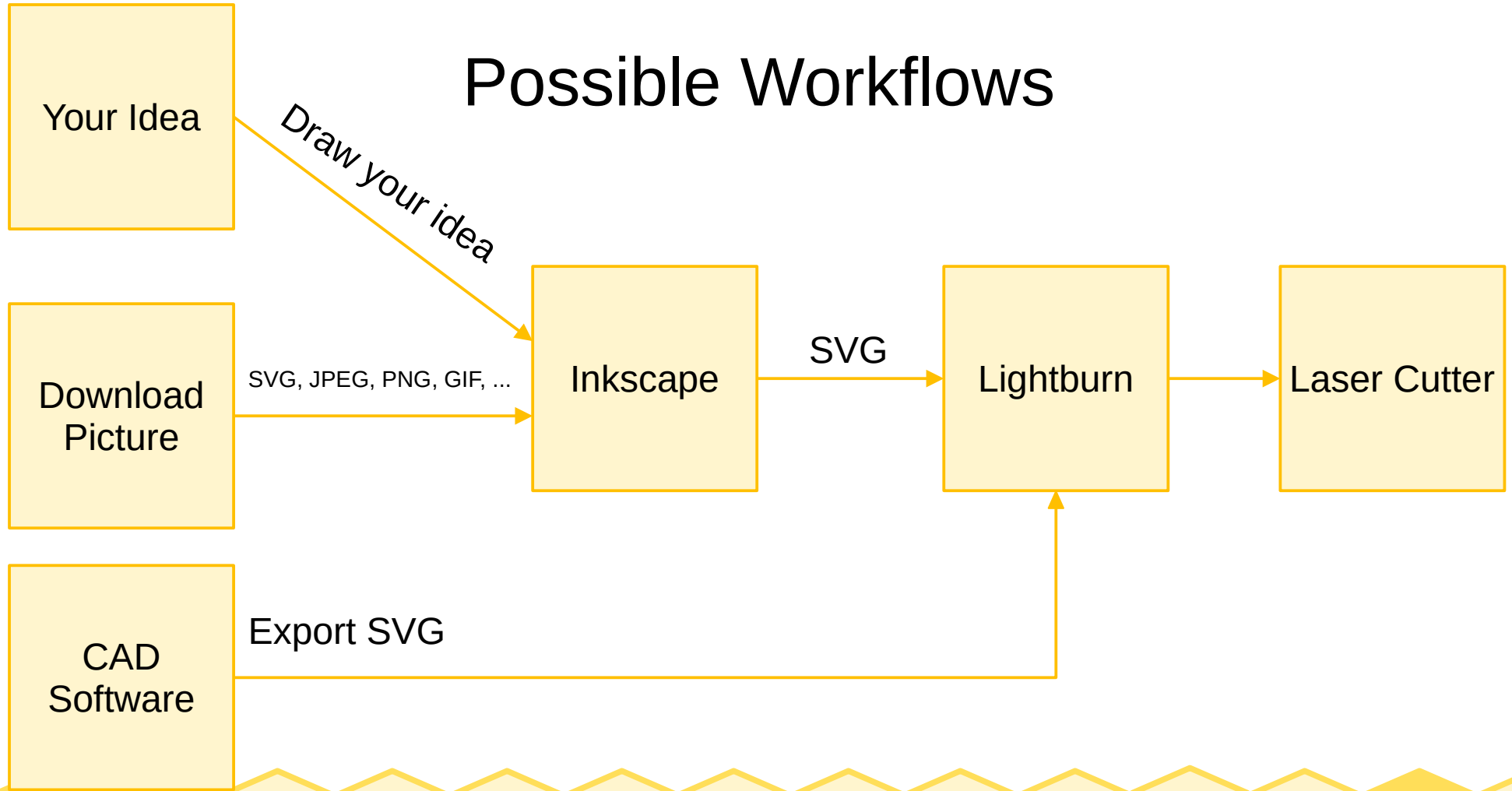


Topics Covered

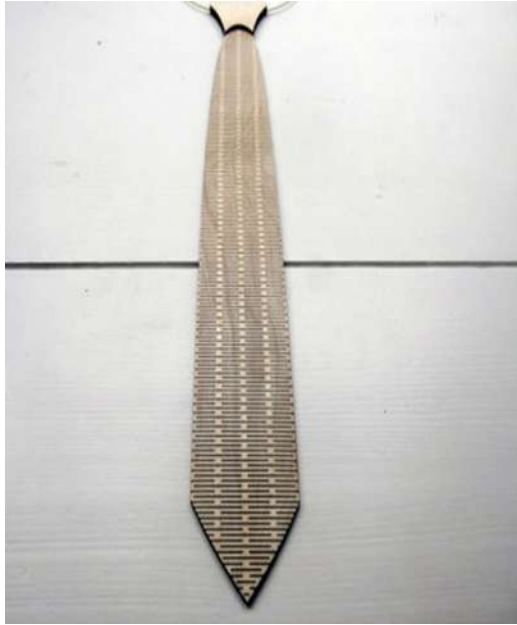
- We will go through a basic design in Inkscape
- We will import file into Lightburn for cutting
- We will set up cutter for operation
- You will laser etch your own cutting board to take home



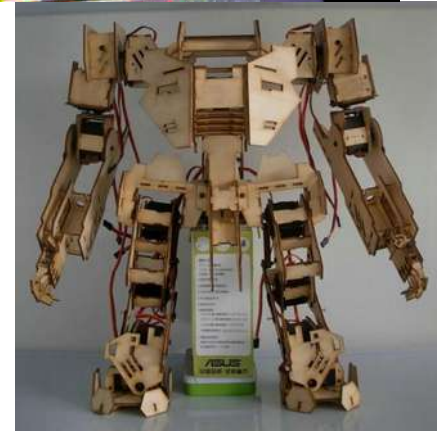
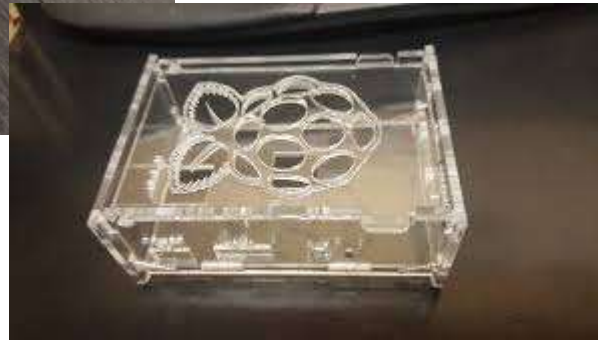
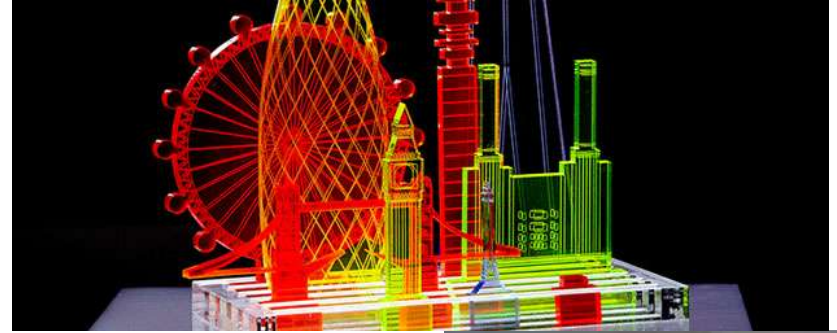
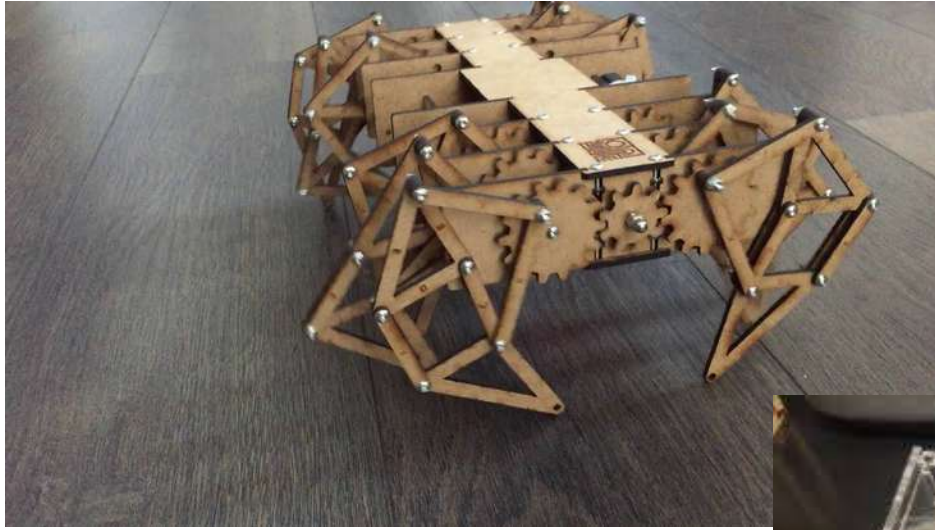
Possible Workflows



Example Projects (Inspiration)



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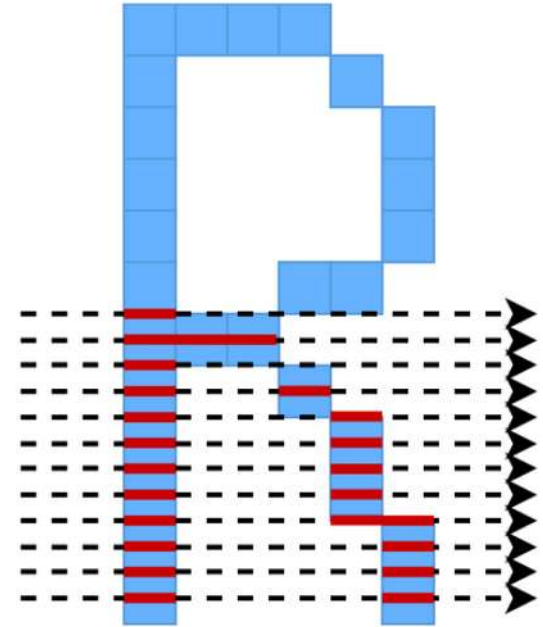
Vector and Raster images

- Vectors are paths that the laser will follow
-
- Raster images are “pixels” or points that will be etched



Cutting raster images

- Raster images will be converted to black and white
- Laser head will “Scan” back and forth pulsing the laser for each dot.

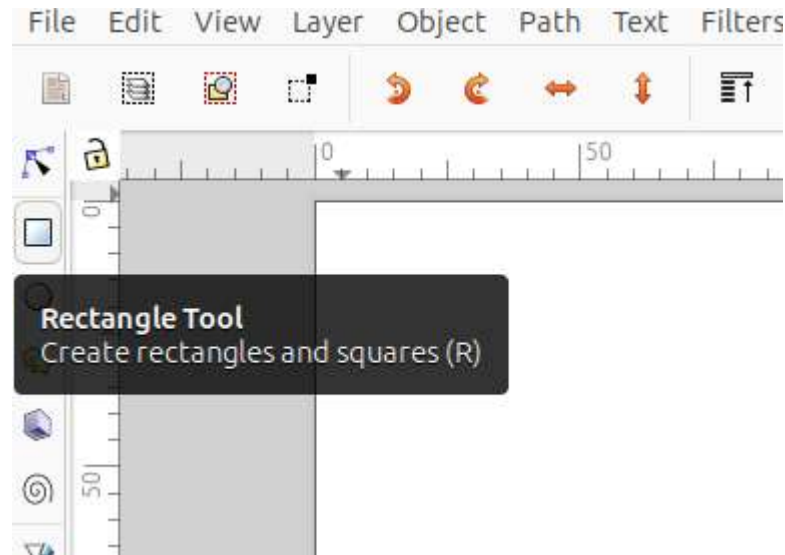


Inkscape



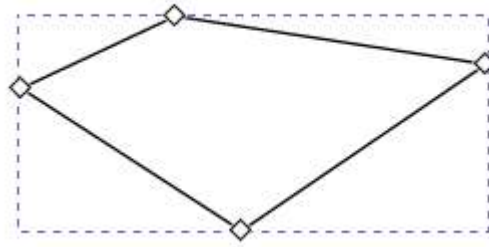
Example design

- Rectangle tool to create rectangles

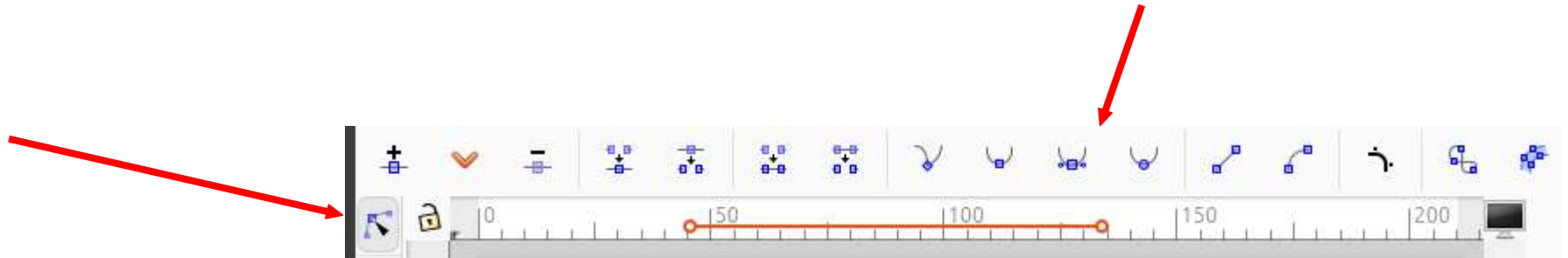


Pencil Tool

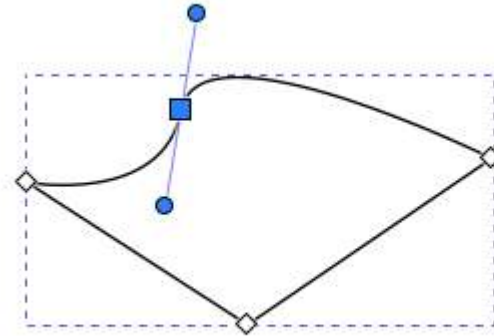
- Pencil tool for straight lines
- Nodes are the endpoints
- You can drag control points to make curves



Nodes



- Adjust node control points to make curves



Fill and Stroke

- Set color in fill and stroke section
- Fill = None for just outline
- Fill color set for “engraving” the inside



Cutting Board Dimensions

- Jig locates board at 60mm to the right and 60mm down from the top left corner of the design
- Board measures 256.4mm wide and 383.4mm tall.
- Decide if you want your board to be portrait or landscape
- Make sure your design fits inside the board
- You can use the example design SVG as a starting point



Finished Design

Design starts at 60mm, 60mm

Fits within the cutting board shape

Leaves a little margin at the edges



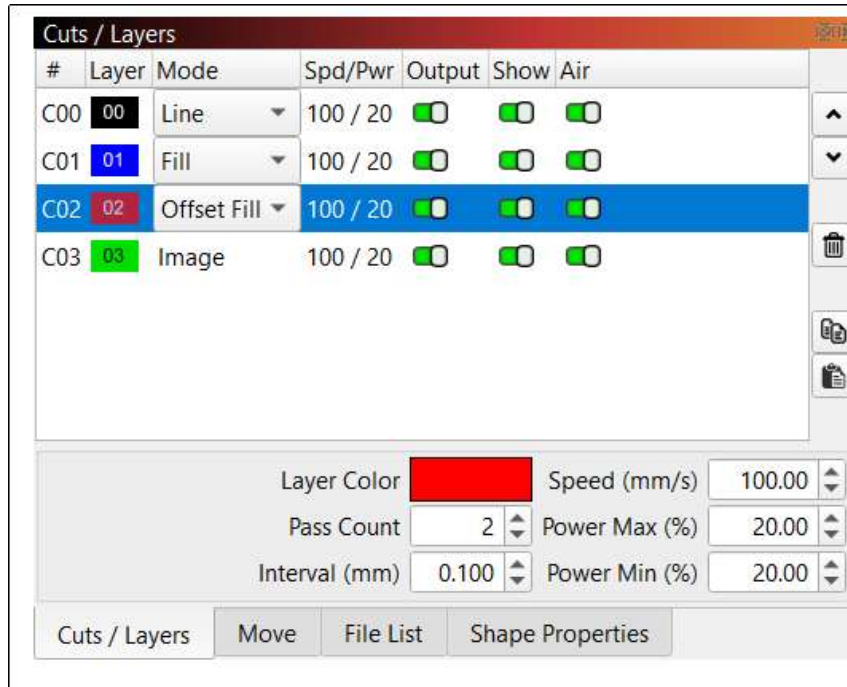
Speed and power

- Each elementy in your design should be given a “color”. That “color” will later determine how the cuts happen.
- Generally, each color will be assigned a “speed” and a “power” level. This will determine how the material gets cut.
- Generally, higher power = more cutting
- Lower speed = more cutting
- Etch/engrave : High speed, low power
- Cut: Low speed, high power (never above 50%)



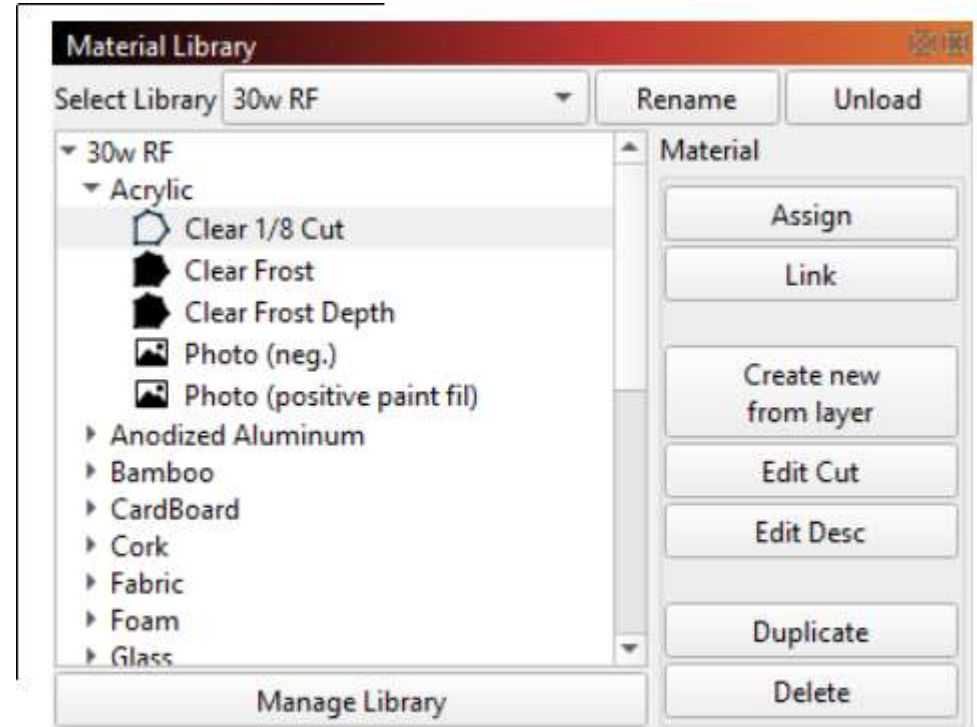
Lightburn

- Import Design
- Set speed and power settings
- Assign each color to a “mode”, and speed/power.
- Never exceed 50% power



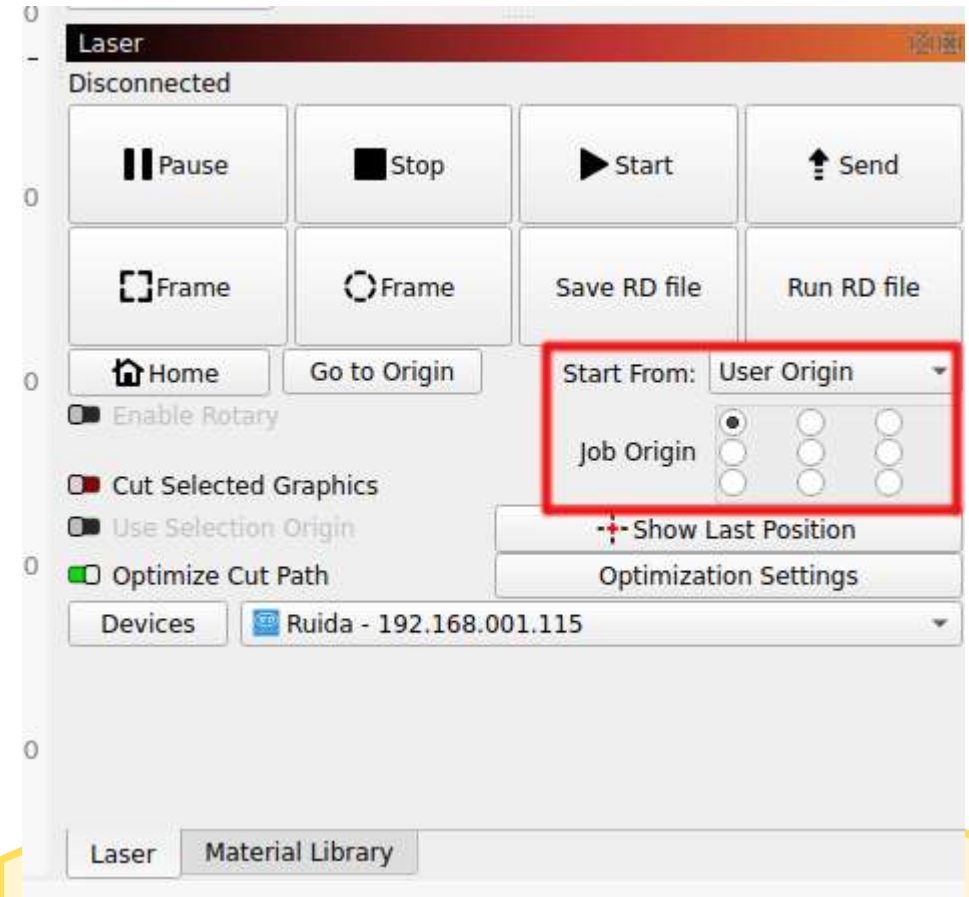
Material Library

- You can set the speed/power manually if you want, or you can “Assign” it from the material library.
- Make sure your “Cut” layer(s) are last, otherwise your work may fall down before it does the engraving layers.



Send to machine for cutting

- Once speed and power are set, we can send our design to the machine to cut it.
- Start button directly starts machine
- Send button sends file to machine. On machine, select file and press start button.



Safety

- Materials: Approved

- MDF
- Plywood and most woods
- Some plastics
- UNTREATED leather
- Stone, Glass, Ceramic

- Materials: Banned

- Plastics containing chlorine
 - Vinyl, PVC, Nylon
- Resins
- Epoxies
- Metals



Safety

- Make sure all panels are closed
- Beware of fire and flare-ups on materials. Be prepared to extinguish fire with fire blanket
- Keep lid closed to avoid blindness
- Beware in case lens vaporizes (unlikely but possible), possible Selenium exposure
- Be careful not to touch lenses or mirrors as they can bring the machine out of alignment.
- Persons under 18 must have parent or guardian nearby



Making the cuts

- Put material in machine
- Your supervisor will turn on the machine, chiller, and air assist as well as opening the air vents.
- Set focal distance between work and laser head.
- Send design to machine
- Close all doors/panels.
- Select File
- Press green “Start/Pause” button.
- Don’t stare at beam
- Don’t leave unattended



Congratulations!

- You have now made your first laser-cut artifact.
- Everything else is creativity and honing your technique

